The New Commute: Realigning BART Schedules to Post-Pandemic Realities

Final Executive Report

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Repository: sjsu-cs133-s26/Group3Riders-BartRidership-transit

Interactive Dashboard: Project_Assignment_Final.ipynb - Colab

Executive Summary

For decades, the Bay Area Rapid Transit (BART) system operated on a predictable rhythm, where we see large numbers of commuters traveling into downtown San Francisco in the morning and leaving in the evening. As a result, we see train schedules, budgets, and maintenance being built around this traditional "rush hour."

However, commuting patterns have changed significantly over the recent years. To better understand this shift, our analysis focuses on two key periods: 2019 as a pre-pandemic baseline and 2022 as a representation of post-pandemic recovery. By analyzing millions of trips across top downtown stations, our analysis revealed three critical shifts:

1. **The Deflation of the Commute Peak:** The massive surges at 8:00 AM and 5:00 PM have collapsed. Stations that used to see 7,000 average riders at 5:00 PM now see under 2,000. This demonstrates how stations who have once experienced heavy surges during peak hours are now operating at a lower volume.
2. **The 5-Day Work Week is Now a 3-Day Work Week:** Weekday travel patterns are no longer consistent across all five days. Tuesdays, Wednesdays, and Thursdays now form a stable "hybrid work" cluster, while Mondays and Fridays have plummeted to resemble weekend levels.

3. **The Collapse of Downtown Micro-Flows:** Transit between the downtown stations (e.g., Civic Center to Embarcadero) has declined sharply, indicating reduced movement between central business districts.

Recommendation: BART should pivot away from a schedule designed primarily around peak-hour commuting and instead focus on an evenly distributed service model that runs throughout the day, reducing service on low traffic days when the demand for commute is low. By evenly spreading out the current resources, BART would better align to current rider behavior while improving cost efficiency.

1. Project Overview and Context

Public transportation relies on the delicate balance of supply and demand. Running too many trains would lead to unnecessary operational cost, while running too few could result in overcrowding and reduce rider satisfaction, which could drive commuters to rely on alternative forms of transport.

This report is designed for transit planners, city officials, and budget analysts who are responsible for making decisions on how to allocate public transportation resources in the coming years. Historically, city planners relied on previous year's data to predict future needs. However, recent events have made these assumptions less reliable.

By analyzing detailed ridership data, we aim to provide a clearer understanding of how commuting patterns have evolved over time and what those changes mean for future planning. Rather than focusing only on total ridership numbers, this study examines when and where people are traveling and allows us to better understand Bay Area commuting patterns and how it has evolved over time.

2. Dataset Introduction

To understand Bay Area transit dynamics, we utilized the **BART Ridership Dataset**, Our Data is provided by Kaggle user Victor Geislinger who claims to have gotten the data

provided directly from BART linking a webpage that requires administrator passwords to access and updating the Data on the 5th of the month.

- **Scale:** The raw dataset is massive, containing tens of millions of rows spanning from 2011 to 2023. To isolate the specific impact of the pandemic on commuter habits, we filtered our analysis specifically to the 2019 and 2022 subsets.
- **Structure:** The data captures transit at an hourly, highly granular level.
- **Key Features:** The core variables include Date, Hour, Origin Station, Destination Station, and Trip Count.

Spanning from 2011 to May 2, 2023, our BART dataset contains approximately 110.7 million complete, null-free records detailing hourly passenger counts between station pairs. Each entry logs the date, hour, origin and destination stations, and the total number of trips tracked by Clipper cards. To conserve space, hourly pairings with zero trips, such as when the system is closed, are excluded. Even with this filtering, the dataset is massive, 2022 alone accounts for over 8.2 million rows, generated by the hourly ridership across up to 2,500 unique route combinations from BART's 50 stations. Ultimately, this comprehensive dataset is an indispensable tool for mapping historical ridership trends, identifying peak travel times, and predicting future passenger demand throughout the network.

Instead of just reporting total ticket sales, this origin-destination structure maps the exact flow of human capital across the Bay Area. It answers not just how many people are riding, but when and where they are going.

BART System Spatial Data (KML): Sourced from the official BART geographic data repository, this file contains the exact geometric coordinates for all Bay Area rail tracks, station entrances, and station centroids. We utilized this spatial data to construct the geographic mapping features in our interactive dashboard.

3. Why This Problem Matters and Why We Chose It

Our team motto is “Mind the Gap in the Data.” As Bay Area residents, we have experienced the shifting transit landscape firsthand. We noticed trains that used to be standing-room-only were now nearly empty, yet they were still arriving on the same aggressive rush-hour schedules.

We chose to study this issue as it has clear real-world implications. BART is currently facing a significant financial shortfall, and decisions about service levels directly affect both its budget and rider experience. By analyzing ridership data, the project aims to highlight where resources may no longer align with demand, indicating a need for adjustment to better match the current usage to help reduce costs and maintain reliable transportation for riders.

4. Methods and Work Across the Semester

To turn millions of rows of raw data into actionable insights, our team moved through a structured analytical pipeline over the course of the semester.

Exploratory Analysis and Early Visualizations

In the initial phases, we filtered the massive dataset down to the most critical locations: the top four downtown San Francisco stations (Civic Center [CIVC], Powell [POWL], Montgomery [MONT], and Embarcadero [EMBR]).

Early exploration confirmed an overall drop in volume. To understand the operational impact, we mapped the morning commute hours (5:00 AM to 10:00 AM) using regression analysis. We discovered that the aggressive 2019 spike in ridership at 8:00 AM had vanished, replaced by a much flatter, lower-volume curve in 2022.

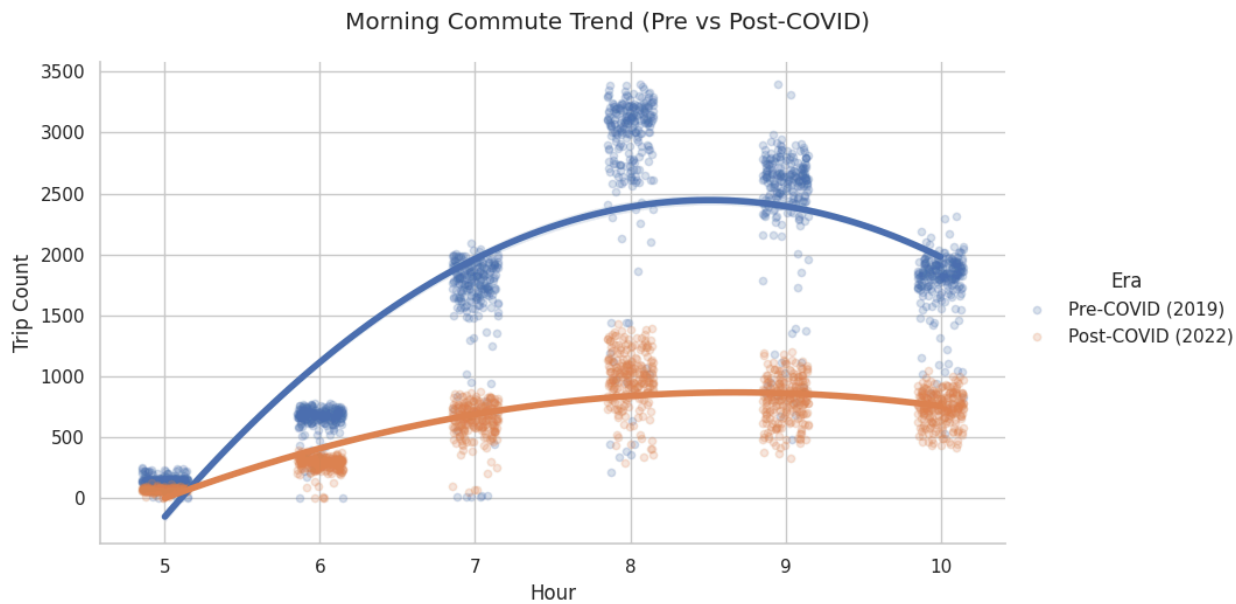


Figure 1: Morning Commute Trend (Pre vs. Post-COVID). A regression analysis illustrating the 5:00 AM to 10:00 AM commuter window.

Intermediate Visualizations and Insights

Next, we needed to know if this pattern varied by station throughout the entire day. We mapped the shape of the day using small-multiples line plots. This answered our primary question: Are people commuting at different times, or are they just not commuting? We learned that the "shape" of the day remains largely the same (a small morning bump and a larger evening bump), but the sheer scale has deflated drastically.

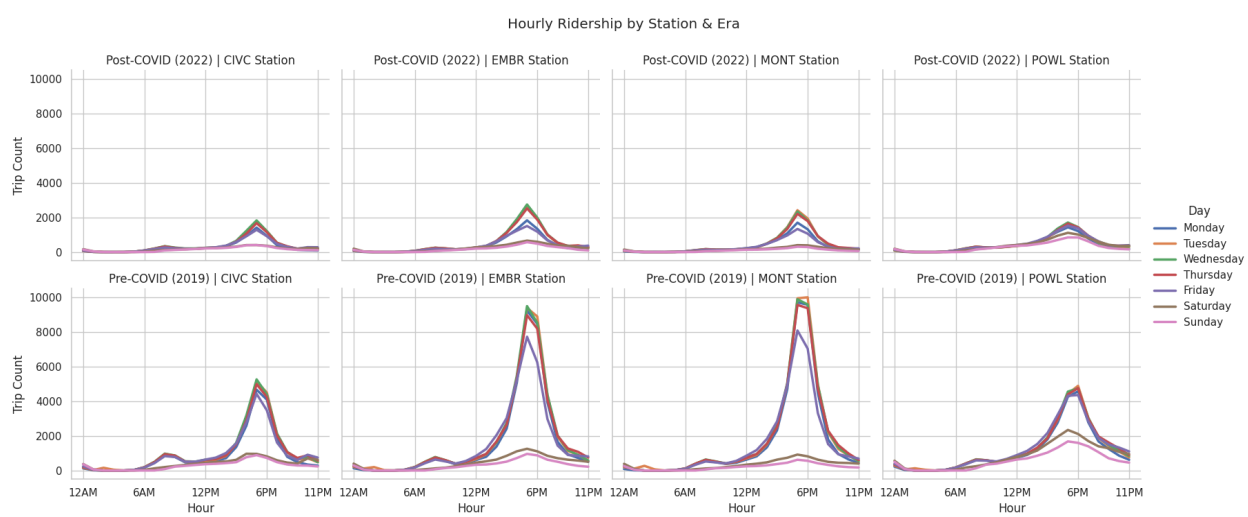


Figure 2: Hourly Ridership by Station & Era. Small-multiples line plots showing the full 24-hour cycle across the four downtown stations.

Dashboard Development

To package our findings for executives, we built an interactive spatial dashboard directly in our Python environment using ipywidgets and folium.

- **What it does:** A user can select any of the stations and toggle between the 2019 and 2022 eras.
- **What it shows:** The dashboard updates a geographic map side-by-side with a detailed bar chart showing average hourly ridership. By locking the Y-axis scale across eras, the dashboard allows decision-makers to toggle between years and physically watch the rider volume collapse.

Deeper Visualizations

To finalize our analysis, we added new, meaningful visualizations to help the data speak clearly to executives regarding flow and weekly predictability:

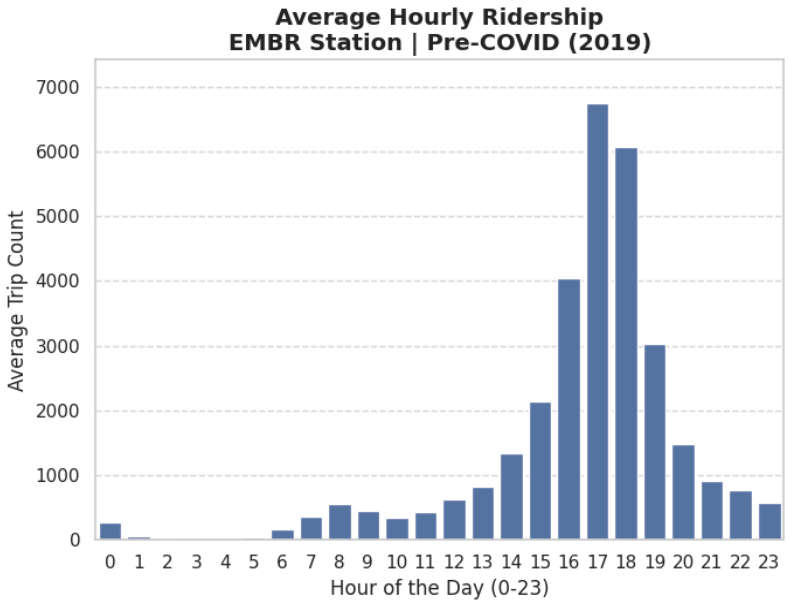
1. **Improved Categorical Box Plot:** We overhauled our daily ridership distribution analysis, applying color shading and explicit text annotations to highlight the new "3-Day Work Week."
2. **Origin-Destination Flow Matrix:** We engineered a dual heatmap with a shared global color scale to visualize exactly how traffic between the four downtown stations collapsed.

5. Key Findings & Recommendations

Based on the synthesis of our historical data analysis, we present the following key findings and strategic recommendations:

Finding 1: The Deflation of the Commute Peak

- **The Insight:** As seen in our Interactive Dashboard outputs, the 5:00 PM (17:00) peak still exists conceptually, but it operates at a fraction of its former capacity. In 2019, Embarcadero (EMBR) saw average peaks of nearly 7,000 trips at 5:00 PM. By 2022, those exact same peaks had deflated to under 2,000 trips.
- **The "So What?":** Maintaining a dense concentration of train departures and station personnel geared toward moving 7,000 people per hour leads to drastically underutilized resources.
- **Recommendation:** Transition from "rush-hour surging" to an evenly spaced train frequency throughout the entire day. Peak-hour frequency should be scaled back to match the massive reduction in actual rider volume.



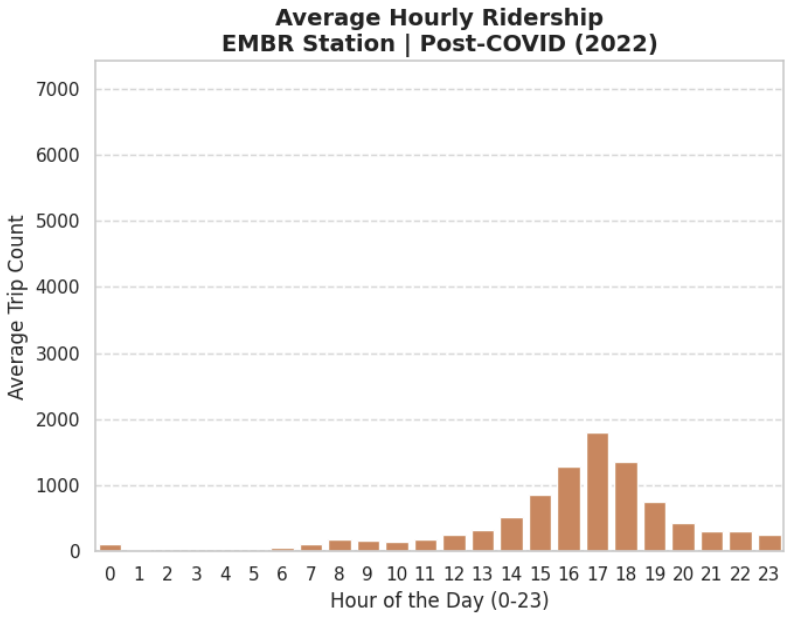


Figure 3: Interactive Dashboard Output (Embarcadero Station). A side-by-side comparison of average hourly ridership.

Finding 2: The 5-Day Work Week is Now a 3-Day Work Week

- **The Insight:** As shown in Figure 2, the traditional Monday-Friday block of consistent transit usage no longer exists. In 2022, Tuesday, Wednesday, and Thursday form a flat "hybrid-work" cluster. Meanwhile, Mondays and Fridays have decoupled from the core work week, dropping heavily and blurring the lines between weekday and weekend volumes.
- **The "So What?":** Running traditional, high-frequency weekday schedules on Mondays and Fridays burns the budget unnecessarily.
- **Recommendation:** Implement a 3-tier scheduling system: Core Mid-Week (Tue-Thu), Flex-Days (Mon/Fri), and Weekends. Monday and Friday schedules should be permanently downgraded to match their new baseline.

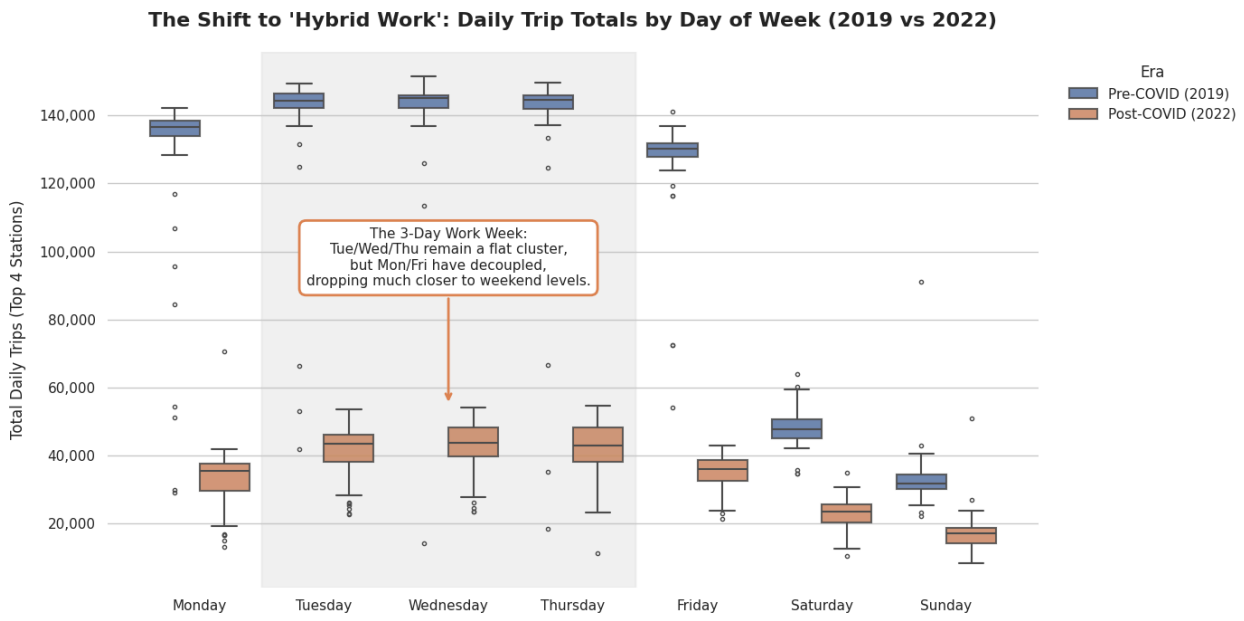


Figure 4: Daily Trip Totals by Day of Week. The annotated region explicitly highlights how Tue-Thu travel forms a new cluster, while Mondays and Fridays have dropped.

Finding 3: The Collapse of Downtown Micro-Flows

- **The Insight:** Transit between the downtown stations has virtually stopped. Looking at the Origin-Destination Heatmap, trips originating at Civic Center (CIVC) and ending at Embarcadero (EMBR) plummeted from ~179,000 total trips in 2019 to just ~43,000 in 2022.
- **The "So What?":** People are no longer using BART for short "micro-commutes" within the downtown financial sector.
- **Recommendation:** Station infrastructure and ticketing gateways at stations like Montgomery and Embarcadero are oversized for current demand. Planners should repurpose extra space in these stations for retail, community space, or alternative revenue generation, as the foot traffic will not return to 2019 levels.

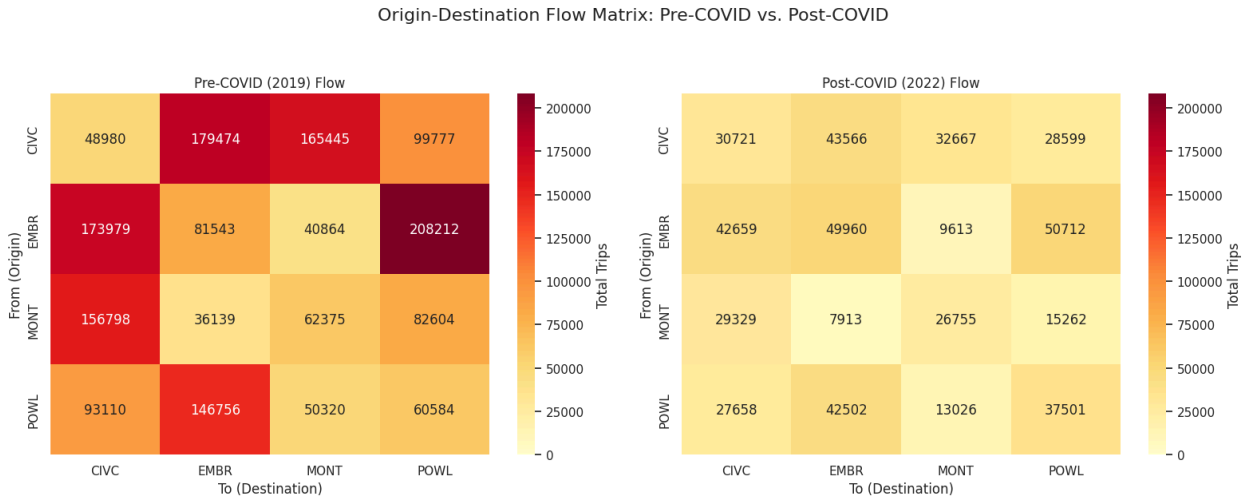


Figure 5: Origin-Destination Flow Matrix. Note the dark red clusters in 2019 indicating massive inter-station travel, which completely fades to light yellow in 2022 due to the shared global color scale.

6. Limitations and Next Steps

While this analysis provides a robust framework for immediate scheduling changes, it does contain limitations:

- **Timeframe Constraints:** Our primary post-COVID insights rely heavily on 2022 data. 2022 was an "early phase" of the new normal where corporate return-to-office mandates were still in flux. It may not perfectly reflect late 2025/2026 realities.
- **Geographic Scope:** To focus on peak commuter zones, our deepest analysis was narrowed to the four downtown San Francisco stations. This may skew our findings toward white-collar corporate work cultures and may not accurately reflect the needs of riders traveling exclusively between suburban stations.
- **Missing Variables:** This dataset strictly measures human movement. It does not account for external variables like weather, holidays, or local sporting events, which likely account for the extreme outliers visible in our box plots.

Next Steps:

If given more time and resources, our next steps would be to merge this dataset with public

event calendars (like Giants/Warriors games) and historical weather data to explain anomalous peaks. Furthermore, we would train a Time-Series Machine Learning model to forecast exact hourly ridership for specific stations, allowing BART to implement a dynamic, AI-driven daily schedule.

Appendix

- **GitHub Repository:** sjsu-cs133-s26/Group3Riders-BartRidership-transit
- **Instructions to run the dashboard:** Open the Google Colab notebook link. Ensure the Kaggle bart-ridership dataset and KML geometry files are loaded in the environment. Run all cells sequentially. The interactive ipywidgets dashboard will render at the bottom of the notebook. Select a station and era from the dropdown menus to dynamically update the Folium map and Seaborn bar charts.
- **Data Source:BART Ridership | Geospatial Data | Bay Area Rapid Transit**